SRN R 2 3 D F 0 4 2

REVA UNIVERSITY
Bengaluru, India

School of Computer Science and Applications

III Semester MCA- IA1 Examination March 2025

Course Code: M23DE0301 | **Duration: 75 Mins**

Course Title: CLOUD COMPUTING AND SERVICES | **Max. Marks: 30**

Q. No	Question	Marks	CO	PO	PSO	BL
	Section A Answer ***any two*** questions.					
1	Discuss *any five* features of Cloud computing.	5	1	1,2,3	1	3
2	Elaborate on the vision of Cloud Computing.	5	1	1,3	1,2	2
3	Explain the Full Virtualization technique.	5	1	1,2	1,2	3
	Section B Answer ***all*** questions.					
	UNIT-I1					
4	a) Briefly summarize the Cloud Computing Reference Model.	2	1	1,3	1,2	1
	b) Discuss the following core technologies that played an important role in the realization of cloud computing: 1. Distributed Systems 2. Web 2.0 3. Service Orientation, and 4. Utility Computing	8	1	2,4	1,2	4
	OR					
5	a) Discuss Scalability and Elasticity with reference to the Cloud Computing.	2	1	1,5	1,2	1
	b) Cloud computing has some interesting characteristics that bring benefits to both cloud service consumers (CSCs) and cloud service providers (CSPs). Explain these characteristics in details.	8	1	1,4	1,2	4
	UNIT-II					
6	a) Mention any two advantages of storage virtualization.	2	2	4	1,2	1
	b) Define virtualization. Describe the characteristics of virtualized environments.	8	2	1,4	1,2	2
	OR					
7	a) List out any two advantages of the Hardware Level Virtualization.	2	2	1,3	1,2	1
	b) Analyze and elucidate the structure of the Machine Reference Model with neat diagram	8	2	1,3	1,2	3

SRN R 23 0 F 0 4 2

REVA UNIVERSITY
Bengaluru, India

School of Computer Science and Applications

III Semester MCA - IA 2 Examination

May 2025

Course Code: M23DE0301	**Duration: 75 Mins**
Course Title: Cloud Computing and Services	**Max. Marks: 30**

Q. No	Question	Marks	CO	PO	PSO	BL
	Section A Answer *any two* questions.					
1	Describe the architecture of the Cloud Reference Model with a neat diagram.	5	3	1,2,3	2	3
2	Write a short note on DataOps and its process with a supporting diagram	5	4	1,2,3	2	3
3	Discuss the features and benefits of Azure Synapse Analytics	5	4	1,2,3	2	3
	Section B Answer *all* questions.					
4	a) Mention any two differences between public and private clouds.	2	3	1,2,3	2	1
	b) Describe Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS) with examples. Compare their characteristics.	8	3	1,2,3	2	3
	OR					
5	a) Write any two advantages of SaaS.	2	3	1,2,3	1,2	4
	b) Explain the reference model of Platform-as-a-Service with a neat diagram.	8	3	1,2,3		3
6	a) Define big data analytics.	2	4	1,2,3	1,2	1
	b) Explain in detail the architecture and workflow of Azure Synapse Analytics. How does it support modern data analytics?	8	4	1,2,3	1,2	3
	OR					
7	a) What is meant by data ingestion in big data processing?	2	4	1,2,3	1,2	2
	b) Discuss the feature of Azure Synapse Analytics.	8	4	1,2,3	1,2	3

*****

Date	16/06/2025
SRN	R23DE042

REVA UNIVERSITY
Bengaluru, India

3rd Semester MCA Semester End Examination June 2025
Course Title: Cloud Computing and Services
Course Code: M23DE0301 - 48623301

Duration: 3 Hours **Max. Marks: 100**

Note:

1. *Answer the required number of questions from **Section A to C.***
2. *Verify and ensure that question paper is completely printed before answering the question paper.*
3. *Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.*
4. ***Students must check the course title and course code before answering the question paper.***

	Section - A	Marks	CO	PO	BL
	Answer any FIVE of the following questions, carrying 2 marks each	**(5x2=10)**			
1.	List the vision introduced by cloud computing. 2		1	1	2
2.	Mention the classification or taxonomy of virtualization at different levels.		2	2	2
3.	List down the characteristics of Cloud Computing. 2		1	1	2
4.	Illustrate community clouds and their typical usage scenarios. 2		3	1,2	2
5.	Interpret one use case of Big Data Analytics.		4	1,2,3	2
6.	Name and briefly describe the three main cloud service models used in cloud computing. 2		3	1,2	2
	Section - B				
	Answer any EIGHT of the following questions, carrying 5 marks each	**(8x5=40)**			
7.	*What is scalability?* Discuss Horizontal Scaling and Vertical Scaling with an example for each.		1	1	2
8.	Compare and contrast distributed systems, virtualization, and utility computing. 5		1	1	4
9.	Describe the characteristics of virtualized environments. 5		2	2	2
10.	Discuss the working of the Xen hypervisor. 5		2	2	2
11.	Explain the concept of utility-oriented computing with an example. 5		1	1	3
12.	Describe the architecture of the Cloud Reference Model. 5		3	1,2	3
13.	Explain the characteristics of public, private, and hybrid clouds. 5		3	1,2	3
14.	Discuss the role of IoT, Machine Learning, and Artificial Intelligence in Big Data Analytics. 2		4	1,2,3	3
15.	Define DataOps. Explain the DataOps process with a neat diagram, describing the role of each component involved.		4	1,2,3	3
16.	Discuss the benefits of combining Power BI and Azure Synapse Analytics? 3		4	1,2,3	3
	Section - C				
	Answer any FIVE of the following questions carrying 10 marks each	**(5x10=50)**			
17.	Discuss the five core technologies that played an important role in the realization of cloud computing. 8		1	1	3
18.	Explain the two types of hypervisors with a neat diagram. 8		2	2	3

19.	Elaborate on different computing platforms and technologies like AWS, Azure, Google AppEngine, Hadoop, and Salesforce.		1	1	3
20.	Explain the architecture and functioning of the Aneka platform. How does it support cloud application development?		3	1,2	3
21.	Define Big Data. Explain the five characteristics of Big Data with suitable examples.		4	1,2,3	3
22.	Discuss the key features and services offered by Microsoft Azure for machine learning. Explain how these features support the end-to-end machine learning lifecycle, including data preparation, model training, deployment, and monitoring.		4	1,2,3	3

***